

Suryaansh Jain

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EDUCATION

UNIVERSITY OF MASSACHUSETTS AMHERST | *M.S. in Computer Science* | GPA: 3.95/4.0 2025 – 2027

- Coursework: Adv. Topics in ML, Reinforcement Learning, Robotics, Industry Research (Adobe), Complexity Theory, Systems for Machine Learning*, Information Retrieval*, Introduction to Simulation*

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD | *B.Tech in Computer Science* | GPA: 9.41/10 2021 – 2025

- Selected coursework: Foundations of ML, Deep Learning, Adv. Topics in DL, RL, Computer Vision, Convex Optimization, Information Theory, Spectral Graph Theory, Game Theory, Theory of Computation, Computational Complexity, Approximate Algorithms, Cryptology, Quantum Computing

PREPRINTS

Universal-LeWM: A Zero-Shot Generalizable World Model 2026

S. Jain, T. Kulkarni. Manuscript in preparation [†]

Zero-shot Fine-grained Image Captioning with VLAs 2026

S. Jain, V. Garimella, R. Rossi, F. Dernoncourt, J. Wang, K. Goswami, N. Lipka, P. Mathur, S. Yoon. Manuscript being submitted to **AAAI 2027**; first author. [†]

Fantastic ϕ 's and where to find them: Reward Based Abstractions for STAR 2026

S. Jain, S. Chaudhari, N. Vlassis, P. S. Thomas. Manuscript being submitted to **AAAI 2027**; first author. [†]

Beyond Consensus: Mitigating the Agreeableness Bias in LLM Judge Evaluations Oct. 2025

S. Jain, U. Z. Ahmed, S. Sahai, B. Leong. [[arXiv:2510.11822](#)]

A Bound for the Cops and Robber Problem via k -Component Order Connectivity Dec. 2024

S. Jain et al. [[arXiv:2412.02511](#)]

RESEARCH EXPERIENCE

UMASS HUMAN-CENTERED ROBOTICS LAB | *Graduate Researcher* Amherst, MA

Zero-Shot Generalizable Multi-Task World Models Feb. 2026 – present

- Built single world model jointly trained on **71 visual control tasks** (OGBench, Meta-World, RoboDesk; action spaces 2–105 dim) over a frozen **DINOv2** backbone with a universal action encoder (zero-pad + binary mask).
- Added dual natural-language task/robot conditioning via frozen **MiniLM** sentence embeddings, enabling zero-shot transfer to new tasks/embodiments without retraining; matched LeWM single-task specialist on cube within 3 SR points at $\sim 18\times$ less compute.

UMASS HUMAN-CENTERED ROBOTICS LAB | *Graduate Researcher* Amherst, MA

VLA Models for Multi-Robot Formation Control Jan. 2026 – present

- Multi-robot coordination using **VLA models** for language- and vision-conditioned formation control; trained in **NVIDIA Isaac Lab**.

ADOBE RESEARCH | *Student Researcher* San Jose, CA (Remote)

Fine-Grained Image Captioning via RL with Detailed Rewards Jan. 2026 – present

- Designed object-aware perception loss extending **PAPO** via **YOLOv13** per-object **KL terms**.
- Designed **SimLoss**, an information theory inspired loss function to enable **self-supervised fine tuning of VLMs** (Qwen, LLaVA).
- Matched the performance of multi pass methods with $10\times$ improvement in cost and latency.

UNIVERSITY OF MASSACHUSETTS AMHERST | *Graduate Researcher* Amherst, MA

Fantastic ϕ 's and where to find them: Reward Based Abstractions for STAR Sep. 2025 – present

- Designed ϕ -function state abstractions for low variance and bias off-policy estimation. **Beats IS and Model based methods** in all RL environments.

NATIONAL UNIVERSITY OF SINGAPORE | *Visiting Researcher* Singapore

Beyond Consensus: LLM-Judge Bias May 2024 – Oct. 2025

- Quantified **agreeableness bias** across 14 SOTA LLMs ($\sim 96\%$ TPR / $<25\%$ TNR on 366 Python programs); proposed minority-veto + regression calibration, reducing max error to **1.2%** ($2\times$ over best ensemble).

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD Undergraduate Researcher	Hyderabad, India
Cop Number via k -Component Order Connectivity	2023 – 2024
Strengthened state-of-the-art upper bound on cop number from $U /2 + 1$ to $U /3 + 9$; first-author manuscript.	
INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD Undergraduate Researcher	Hyderabad, India
Inter-Blockchain Communication (OTEx)	2023
Worked on OTEx: Ownership Transfer and Execution Protocol for Blockchain Interoperability; Toyota-funded cross-chain research.	

INDUSTRY EXPERIENCE	
APPLIED MATERIALS Machine Learning Intern	Santa Clara, CA
SPG Team	Summer 2026 12 Weeks (Ongoing)
Building time series forecasting models to predict the inventory using Temporal Fusion Transformers and N-HiTS models.	
CROW CANYON SOFTWARE Machine Learning Intern	Remote
Nitro Studio AI Team	Winter 2025 6 weeks
Engineered Agentic RAG systems with LangGraph, leveraging custom agent workflows and PDF-based knowledge retrieval to build customer support chatbots for Nitro Help Desk.	
BRYT SCHOOLS ML & Software Engineering Intern	Remote
Tutor Development Team	Summer 2025 10 weeks
Shipped LLM-powered conversational chatbot: fine-tuning, eval, production deployment across 100+ files.	

ACADEMIC PROJECTS	
Robustness of Object Detectors to Adversarial Attacks	Computer Vision course
	Spring 2025
Reproduced patch- and gradient-based attacks; designed novel variant Edge-DPAttacks on YOLO and Faster R-CNN.	
Diffusion + GNNs for Graph Coloring	Adv. Topics in DL course
	Fall 2024
Diffusion-based generative framework over graphs; 10% error rate on 4-colorable graphs, extended to arbitrary k-colorable.	
Recommender Systems Using Deep Generative Models	Deep Learning course
	Spring 2024
Surveyed SOTA recommender system models like DiffRec, Diff4Rec, and Contrastive Variational AutoEncoder for Sequential Recommendation. Presented these papers and proposed potential improvements.	
Drone Motor Failure Detection	Inter-IIT Tech Meet
	2024
LSTM detector for motor failure on drone-motor telemetry; deployed in C++ within ROS2 for real-time inference on PX4 hardware.	

HONORS & AWARDS	
Selected for Y Combinator Startup School	2026
OnePlus-Oppo Genius+ Scholarship – 6 students nationwide	2022
All India Rank 311 in IIT-JEE Advanced (155K candidates) AIR 458 in IIT-JEE Mains (1M candidates)	2021
Selected for the Indian Chemistry Olympiad Orientation Camp (AIR 27, HBCSE)	2021
KVPY Fellowship, Government of India – awarded twice	2019, 2020

TEACHING & ACTIVITIES	
Teaching Assistant, Reinforcement Learning, UMass Amherst	2026
Teaching Assistant, Information Theory and Theory of Computation, IIT Hyderabad	2024, 2025
3rd Place (2024) and 2nd Place (2023), Math Bowl (CMI) – Inter-IIT Tech Meet	2023, 2024
Head, Web Dev Team, Milan – IIT Hyderabad’s annual cultural festival	2024

TECHNICAL SKILLS	
Languages	Python, C/C++, Java, JavaScript, Bash, SQL, Rust, Dart, LaTeX
ML & Systems	PyTorch, TensorFlow, JAX, OpenCV, CUDA, Docker, ROS2, Gazebo, NVIDIA Isaac Lab, MuJoCo, NumPy, Pandas, scikit-learn, SciPy, Diffusers, CLIP, LangChain, LangGraph, Transformers, Trl, Unsloth, OpenRouter, FastAPI, Django, React, Flask, Flutter, SwiftUI, AWS, AWS Bedrock, OpenAI

† Manuscript available upon request. * Courses in Fall 2026.